

EXP.NO: 5	CPU Scheduling
DATE:	

AIM:

To implement different CPO scheduling calgorithm (FCFS, SJF, Priority cand Round Robin) cand evaluate their performance by calculating Average Jurnaround Time and Average Maiting time.

ALGORITHM:

1. nitialization: Create a structure/class for Processes containing PID, AT, BT, and Priority.
2. Time Tracking: Maintain a global currentTime variable.
3. The Selection Loop: While processes remain:
4. Identify all processes that have arrived ($AT \leq \text{currentTime}$).
5. Apply the specific Selection Criteria (e.g., Min BT, Max Priority).
6. Update the CPU clock and mark completion.
7. Calculation: Compute averages for TAT and WT across all processes.

CODE:

FCFS

```
#include <stdio.h>
int main() {
    int n, ct = 0;
    float tat = 0, wt = 0;
    printf("Enter number of processes:\n");
    scanf("%d", &n);

    for (int i = 0, at, bt; i < n; i++) {
        printf("Enter AT and BT for P%d:\n", i + 1);
        scanf("%d %d", &at, &bt);

        if (ct < at) ct = at;
        ct += bt;

        tat += (ct - at);
        wt += (ct - at - bt);
    }
}
```

```

printf("\nAvg TAT = %.2f\nAvg WT = %.2f\n", tat/n, wt/n);
return 0;
}

```

SJF

```

#include <stdio.h>

int main() {
    int n, ct[20], tat[20], wt[20], at[20], bt[20], done[20] = {0};
    printf("Enter number of processes:\n");
    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        printf("Enter AT and BT for P%d:\n", i + 1);
        scanf("%d %d", &at[i], &bt[i]);
    }

    int time = 0, completed = 0;

    while (completed < n) {
        int idx = -1;
        for (int i = 0; i < n; i++) {
            if (!done[i] && at[i] <= time &&
                (idx == -1 || bt[i] < bt[idx]))
                idx = i;
        }

        if (idx == -1) { time++; continue; }

        time += bt[idx];
        ct[idx] = time;
        tat[idx] = ct[idx] - at[idx];
        wt[idx] = tat[idx] - bt[idx];

        done[idx] = 1;
        completed++;
    }
}

```

```

float avg_tat = 0, avg_wt = 0;
for (int i = 0; i < n; i++) {
    avg_tat += tat[i];
    avg_wt += wt[i];
}

printf("\nAvg TAT = %.2f\nAvg WT = %.2f\n", avg_tat / n, avg_wt / n);
return 0;
}

```

PS

```

#include <stdio.h>

int main() {
    int n, t = 0, c = 0, idx;
    float tw = 0, tt = 0;

    scanf("%d", &n);
    int at[n], bt[n], pr[n], d[n];

    for (int i = 0; i < n; i++) {
        scanf("%d%d%d", &at[i], &bt[i], &pr[i]);
        d[i] = 0;
    }

    while (c < n) {
        int best = 9999;
        idx = -1;

        for (int i = 0; i < n; i++)
            if (!d[i] && at[i] <= t && pr[i] < best)
                best = pr[i], idx = i;

        if (idx == -1) { t++; continue; }

        tw += t - at[idx];
        t += bt[idx];
        tt += t - at[idx];
    }
}

```

```

        d[idx] = 1;
        c++;
    }

    printf("Avg TAT = %.2f\nAvg WT = %.2f", tt/n, tw/n);
}

```

RR

```

#include <stdio.h>
int main() {
    int n, tq, time = 0, done = 0;
    scanf("%d %d", &n, &tq);

    int at[n], bt[n], rem[n], ct[n];
    for (int i = 0; i < n; i++) {
        scanf("%d %d", &at[i], &bt[i]);
        rem[i] = bt[i];
    }

    while (done < n) {
        int flag = 0;
        for (int i = 0; i < n; i++) {
            if (rem[i] > 0 && at[i] <= time) {
                flag = 1;
                if (rem[i] <= tq) {
                    time += rem[i];
                    ct[i] = time;
                    rem[i] = 0;
                    done++;
                } else {
                    time += tq;
                    rem[i] -= tq;
                }
            }
        }
        if (!flag) time++;
    }
}

```

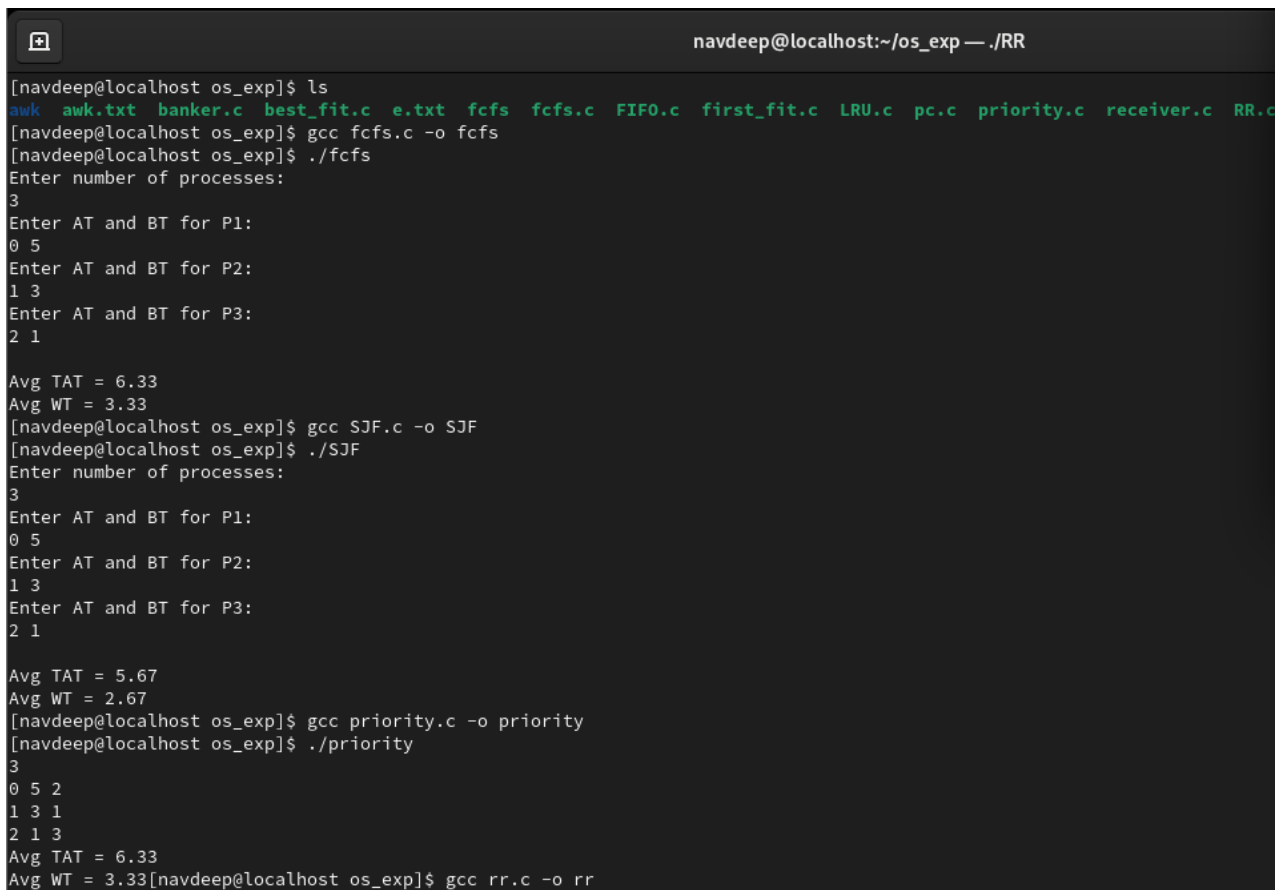
```

float tat = 0, wt = 0;
for (int i = 0; i < n; i++) {
    tat += ct[i] - at[i];
    wt += ct[i] - at[i] - bt[i];
}

printf("Avg TAT=%.2f\nAvg WT=%.2f", tat/n, wt/n);
}

```

OUTPUT:



```

navdeep@localhost:~/os_exp — ./RR
[navdeep@localhost os_exp]$ ls
awk  awk.txt  banker.c  best_fit.c  e.txt  fcfs  fcfs.c  FIFO.c  first_fit.c  LRU.c  pc.c  priority.c  receiver.c  RR.c
[navdeep@localhost os_exp]$ gcc fcfs.c -o fcfs
[navdeep@localhost os_exp]$ ./fcfs
Enter number of processes:
3
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 3
Enter AT and BT for P3:
2 1

Avg TAT = 6.33
Avg WT = 3.33
[navdeep@localhost os_exp]$ gcc SJF.c -o SJF
[navdeep@localhost os_exp]$ ./SJF
Enter number of processes:
3
Enter AT and BT for P1:
0 5
Enter AT and BT for P2:
1 3
Enter AT and BT for P3:
2 1

Avg TAT = 5.67
Avg WT = 2.67
[navdeep@localhost os_exp]$ gcc priority.c -o priority
[navdeep@localhost os_exp]$ ./priority
3
0 5 2
1 3 1
2 1 3
Avg TAT = 6.33
Avg WT = 3.33[navdeep@localhost os_exp]$ gcc rr.c -o rr

```

RESULT:

The implementation confirms that **SJF** is the mathematically "optimal" algorithm for minimizing wait times. However, in modern operating systems, **Round Robin** or a **Multilevel Feedback Queue** is preferred because they prevent "starvation" (where a long process never gets to run) and provide better responsiveness for the user.